rusEFI STI 2004: plug-and-play ECU for the 2004 USDM Subaru Impreza WRX STI (EJ257, drive-by-wire).

Built from Hellen-One modules. Signals between sheets are global net labels; power symbols are global.
Crossed-out (DNP) parts are optional footprints, not assembled: what each one is for is in documents/README.md.
Car pinout and cross-check: documents/STI2004-pinout-check.xlsx (tools/pinout_check.py).
Sheet order follows the signal: car connector -> power -> MCU -> inputs -> outputs -> comms -> aux.

Vehicle connector

File: Connector.kicad_sch

Units: D = B137, C = B136, B = B135, A = B134. Every pin carries a global net label used on the other sheets.

Power

File: Power.kicad_sch

M14 (main power module): +12V_RAW from the main relay (B5/B6), +12V_IGN (D15), back-up +12V (B19).

MCU and USB

File: Mcu.kicad_sch

M1: Hellen mega-mcu144 (STM32F7), with a 120R CAN terminator built in.

Sensors

File: Sensors.kicad_sch

Crank: 2-wire VR sensor (B10 +, B22 -) into M5. Knock (C25) into M13. M7 is a spare knock channel (not used).

Switch inputs

File: Inputs.kicad_sch

M11/M12 (dinput4): car lines that close to ground (defogger D11, lights D12, blower D13, test mode D14,

Outputs

File: Outputs.kicad_sch

Low-side drivers (VND14NV04) in sub-sheets: INJ injectors 1-4; LS1 main relay, fans, ETB relay;

Comms

File: Comms.kicad_sch

M3: rusEFI wideband (WBO) for a Bosch LSU 4.9 front O2, on CAN to M1. It reaches the sensor through the stock

Aux and expansion

File: Aux.kicad_sch

J3: Molex Mini-Fit Jr 2x7 (pinout in documents/README.md): fused 12 V and 5 V, sensor ground, three digital

Sheet: /
File: rusefiSTI2004.kicad_sch

Title:

Size: A3	Date: 2025-02-17	**Rev:**
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Vehicle connector J1 (TE 1-1376430-0): the stock 2004 STI ECM plugs.

Units: D = B137, C = B136, B = B135, A = B134. Every pin carries a global net label used on the other sheets.
Pin functions: documents/oem-ecm-pinout-2004-sti.csv (FSM EN(STi)-18..22, wiring WI-106..118); check: tools/pinout_check.py.
Coil outputs B15-B18 run through F1-F4 (50 mA PTC). R1/R2 divide battery voltage for IN_VMAIN.

J1A
1-1376430-0

D: B137

1 GND_CONTROL_SYS → GND
2
3 GND_ET_SENSOR → GND
4 OUT_ETB-
5 OUT_ETB+
6 → +12V_ETB
7 → GND
8 IN_STARTER_SWITCH
9 IN_NEUTRAL_SWITCH
10 IN_PS_PRESSURE_SWITCH
11 IN_DEFOG
12 IN_SMALL_LIGHT
13 IN_BLOWER
14 IN_TEST_MODE
15 → +12V_IGN
16 IN_AC_REQUEST
17 OUT_MAIN_RELAY
18
19
20 K_LINE
21
22
23
24 IN_BLOWBY
25 IN_O2S
26
27
28 FP_STATUS
29 DLC6_D29
30
31 SHIELD → GND

J1B
1-1376430-0

C: B136

32
33
34 OUT_INJ4
35 OUT_INJ3
36 OUT_INJ2
37 OUT_INJ1
38 IN_CRUISE_MAIN
39 IN_BRAKE_SW2
40 IN_BRAKE_SW1
41 IN_CRUISE_RESUME
42 IN_CRUISE_SET
43
44 IN_IAT
45 IN_ECT
46 APS_PS_5V
47 → +5VAS
48 IN_PPS_MAIN
49 IN_TPS_MAIN
50
51 IN_FUEL_LEVEL
52
53 IN_MAP
54 IN_MAF
55
56 IN_KNOCK_RAW
57 IN_TUMBLE_SENSOR_LH
58 IN_TUMBLE_SENSOR_RH
59 IN_PPS_SUB
60 IN_TPS_SUB
61
62 GND_SENSOR
63 MAS_SHIELD
64 KNOCK_SHIELD → GNDGND
65 GND_SENSOR
66 GND_SENSOR

J1C
1-1376430-0

B: B135

67 GND_PWER_SUPPLY → GND
68 OUT_O2_HEATER_REAR
69
70 GND_PWER_SUPPLY → GND
71 → +12V_RAW
72
73 R1 33k → IN_VMAIN
74 IN_CAM
75 IN_CAM_RH
76 IN_CRANK+
77
78 GND_IGN → GND
79
80
81 F1 nSMD005 → OUT_IGN4
82 F2 nSMD005 → OUT_IGN3
83 F3 nSMD005 → OUT_IGN2
84 F4 nSMD005 → OUT_IGN1
85 → +12V_BACK_UP
86
87
88 IN_CRANK-
89
90 OUT_FAN_RELAY_2
91 OUT_FAN_RELAY_1
92 FUEL_PUMP_SIGNAL1
93 IN_VSS
94
95
96
97 SHIELD → GND
98
99 OUT_AC_RELAY
100
101 OUT_ELECTRONIC_THROTTE_RELAY

R2 6.8k → GND

J1D
1-1376430-0

A: B134

102 IN_CLUTCH
103 WBO_HEATER
104 WBO_HEATER
105
106
107 GND_FRONT_02 → GND
108
109 OUT_TGV_CLOSE
110 OUT_TGV_OPEN
111 OUT_TGV_CLOSE
112 OUT_TGV_OPEN
113 EVAP_A12
114 EVAP_A13
115 OUT_PURGE_CONTROL
116 OUT_CRUISE_MAIN_LIGHT
117 OUT_CRUISE_SET_LIGHT
118 OUT_MIL
119 → +12V_RAW
120
121
122
123 CHARGE_A22
124 OUT_TACH
125
126 WBO_IP_J1
127 WBO_VM_J1
128
129 OUT_VVT_RH-
130 OUT_VVT_LH-
131
132
133 OUT_WG
134 WBO_UN_J1
135 → GND

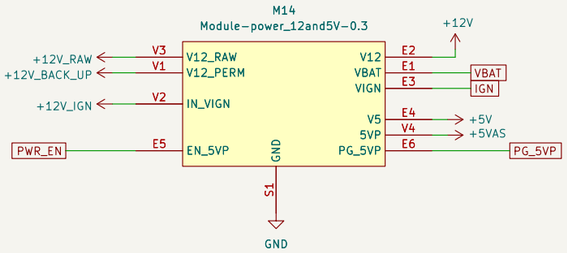
Sheet: /Vehicle connector/
File: Connector.kicad_sch

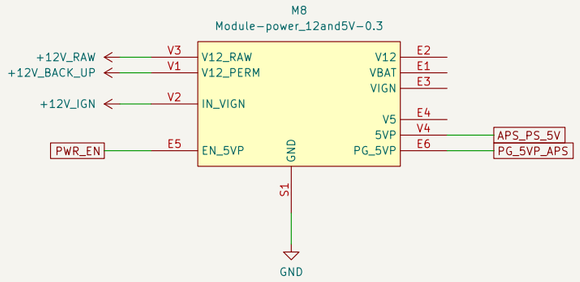
Title: Vehicle connector

Size: A3	Date: 2025-02-17	**Rev:**
KiCad E.D.A. 10.0.7		Id: 2/14

Power.

M14 (main power module): +12V_RAW from the main relay (B5/B6), +12V_IGN (D15), back-up +12V (B19).
It makes +12V, +5V and +5VAS, the protected sensor 5 V for throttle TPS, MAP and TGV sensors (C16).
M8 (second power module) feeds only the pedal 5 V (APS_PS_5V, C15), so pedal and throttle sensors are on independent rails and their plausibility check works. The open link puts the pedal on +5VAS if M8 is not fitted.









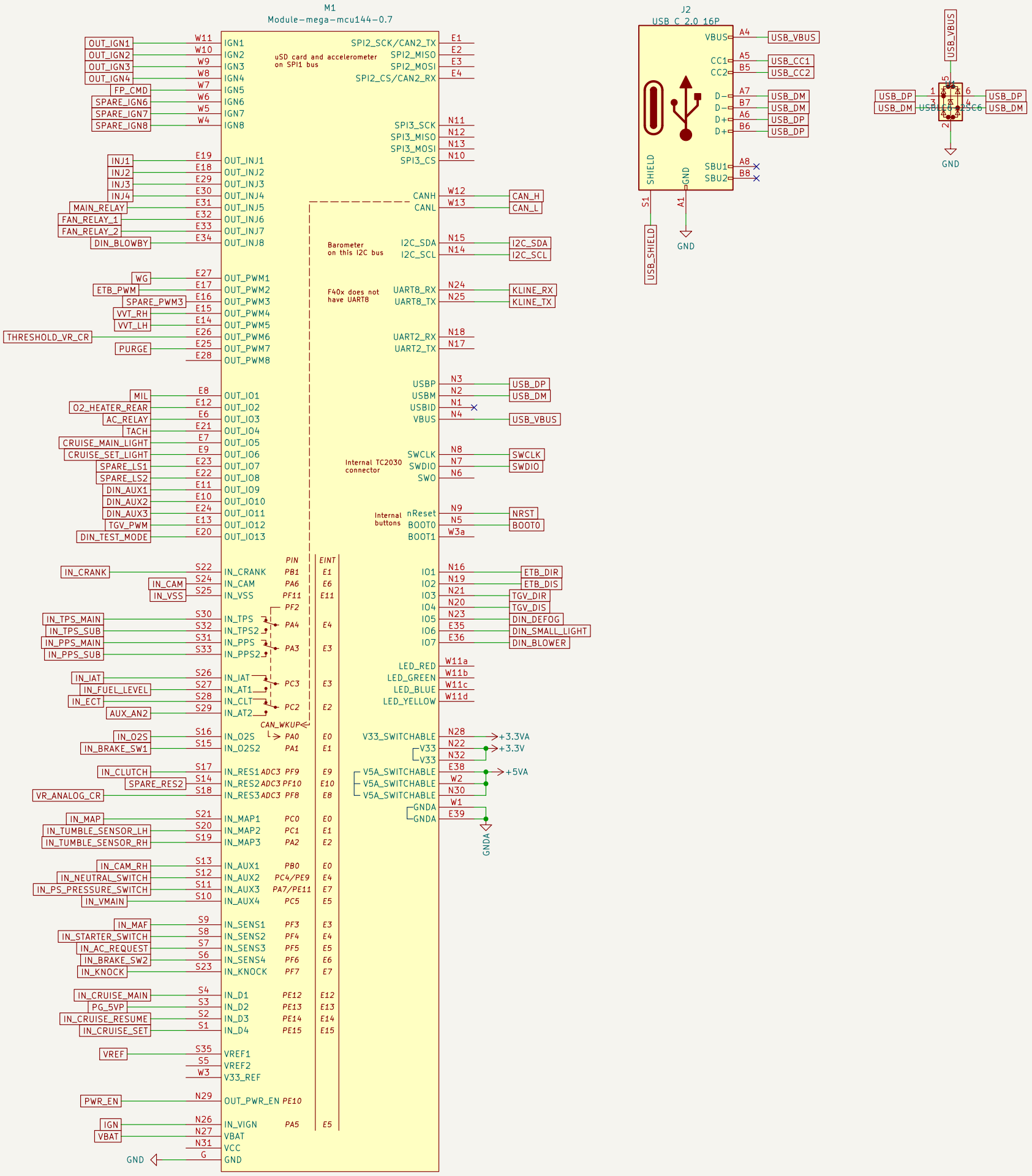
Sheet: /Power/
File: Power.kicad_sch

Title: Power

Size: A3	Date: 2025-02-17	**Rev:**
KiCad E.D.A. 10.0.7		Id: 3/14

MCU and USB.

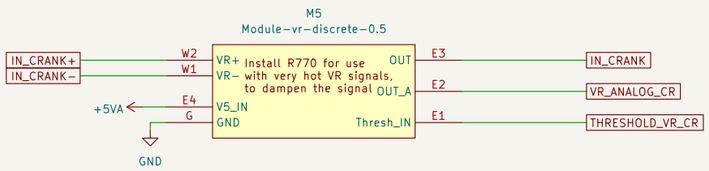
M1: Hellen mega-mcu144 (STM32F7), with a 120R CAN terminator built in.
J2 USB-C for TunerStudio: 5.1k CC pull-downs (device), USBLC6-2SC6 ESD (U1), shell to GND through a 0R link.
BOOT0 pad with an open link to 3.3 V forces the USB DFU bootloader; SWD pads (SWDIO, SWCLK, NRST) for a debugger.

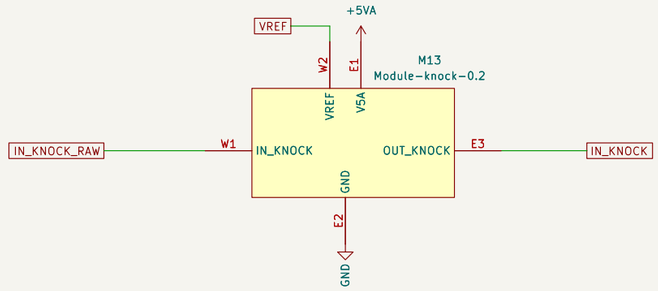


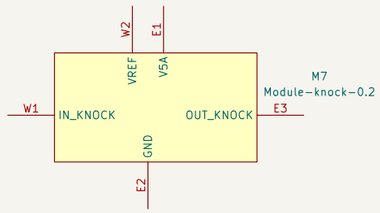
Sheet: /MCU and USB/	
File: Mcu.kicad_sch	
Title: MCU and USB	
Size: A3	Date: 2025-02-17
KiCad E.D.A. 10.0.7	Id: 4/14

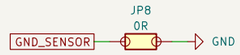
Engine sensors.

Crank: 2-wire VR sensor (B10 +, B22 −) into M5. Knock (C25) into M13. M7 is a spare knock channel (not used).
Cams: 3-wire Hall sensors (WI-114), wired straight to M1 inputs that have pull-ups; crossed parts = stronger pull-ups.
Analog sensors (MAF, MAP, IAT, ECT, TPS, pedal, TGV, fuel level) go straight to M1 inputs by net label.
Sensor returns C31/C34/C35 (GND_SENSOR): to GND through the fitted 0R, or to GNDA through the open link.











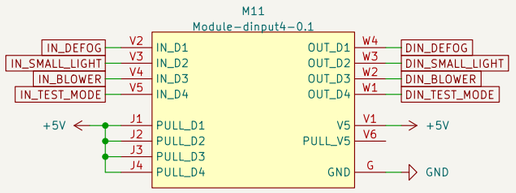


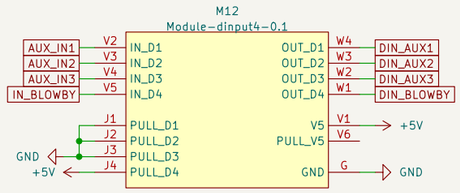


Sheet: /Sensors/		
File: Sensors.kicad_sch		
Title: Sensors		
Size: A3	Date: 2025-02-17	**Rev:**
KiCad E.D.A. 10.0.7		Id: 5/14

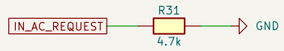
Switch inputs.

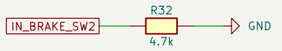
M11/M12 (dinput4): car lines that close to ground (defogger D11, lights D12, blower D13, test mode D14, blow-by D24) with 680k pull-ups, and three aux inputs (launch / 2-step, map switch, spare) that expect 12 V. 12 V-active switches (starter, A/C request, brake 1 and 2, clutch) and the cruise buttons are read on M1 analog inputs with the fitted pull-downs here. Crossed-out parts are options: see documents/README.md.

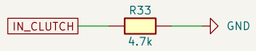


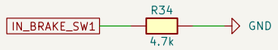


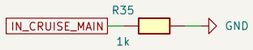


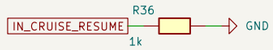


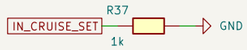




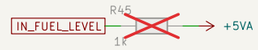














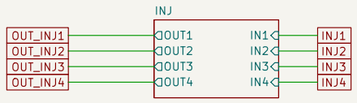


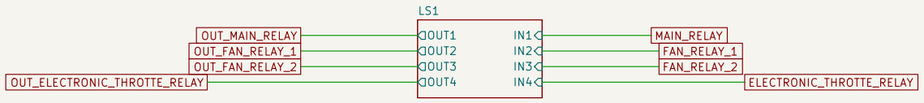


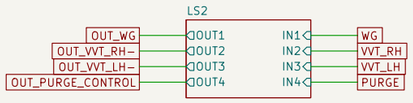
Sheet: /Switch inputs/ File: Inputs.kicad_sch		
Title: Switch inputs		
Size: A3	Date: 2025-02-17	**Rev:**
KiCad E.D.A. 10.0.7		Id: 6/14

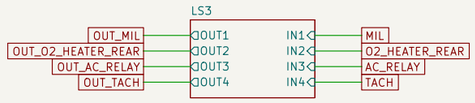
Outputs.

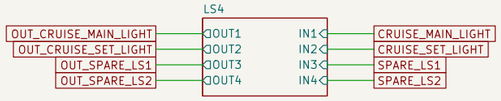
Low-side drivers (VND14NV04) in sub-sheets: INJ injectors 1-4; LS1 main relay, fans, ETB relay;
LS2 wastegate, AVCS RH/LH, purge; LS3 check-engine lamp, rear O2 heater, A/C relay, tach;
LS4 cruise lamps and two spares (aux connector, or EVAP A12/A13 through the open links).
M2 drives the electronic throttle motor; M9 drives both TGV motors in parallel.
Fuel pump control unit command B26: 5 V PWM from IGN5 through 100R, with pull-up/down options.

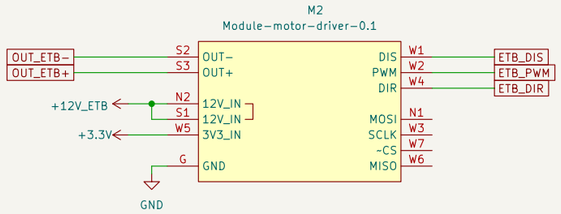


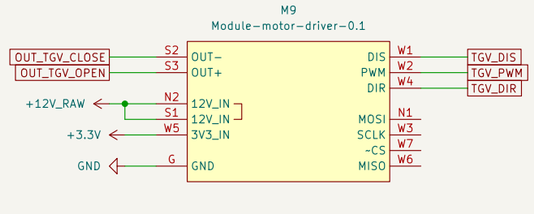


























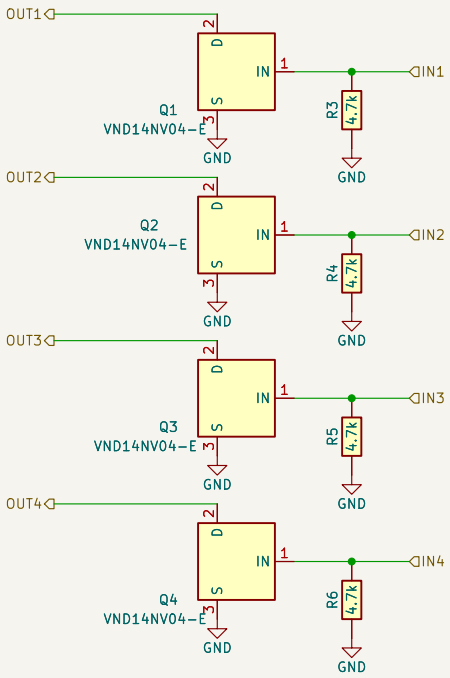




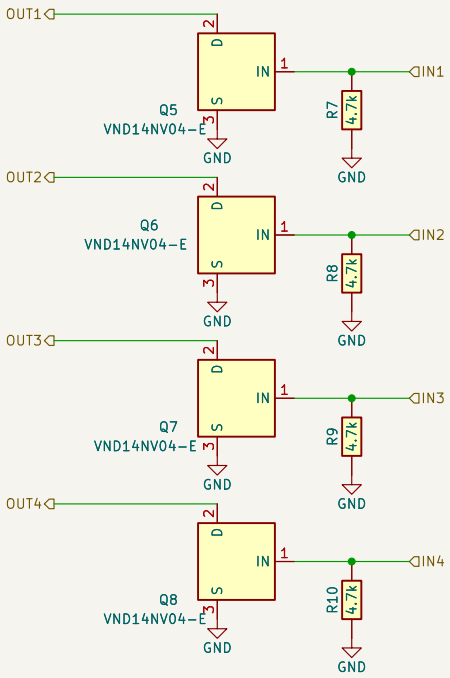
Sheet: /Outputs/
File: Outputs.kicad_sch

Title: Outputs

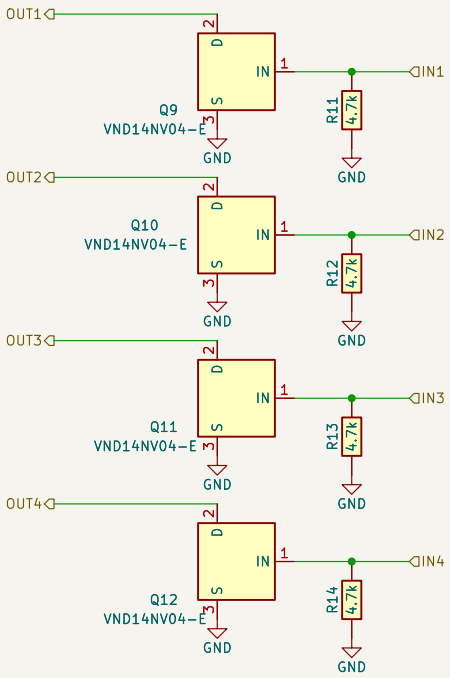
Size: A3	Date: 2025-02-17	**Rev:**
KiCad E.D.A. 10.0.7		Id: 7/14



Sheet: /Outputs/INJ/		
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Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 10/14



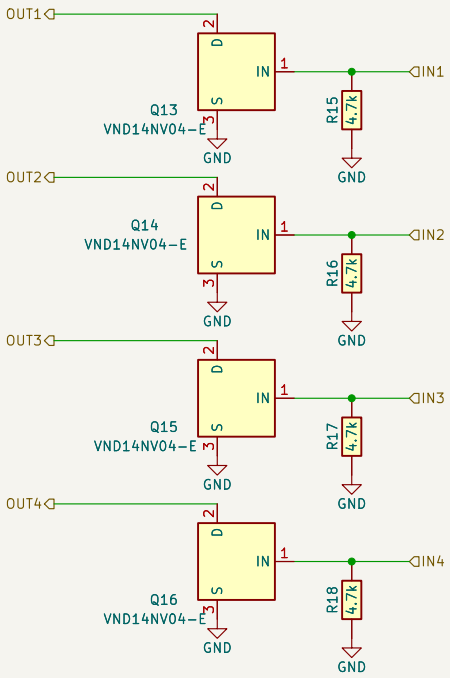
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Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 11/14



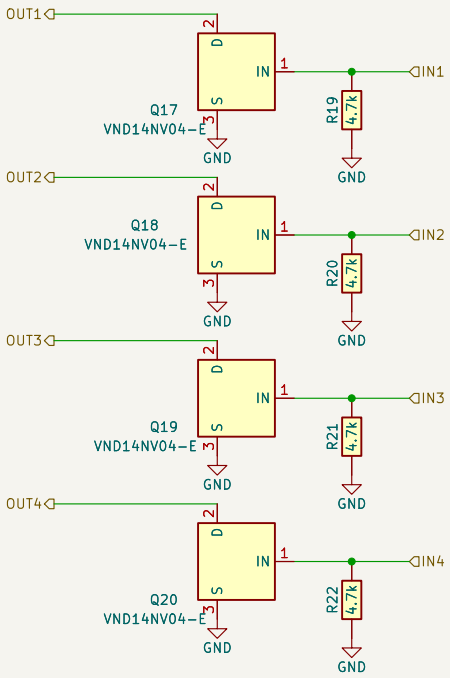
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File: INJ4.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 12/14



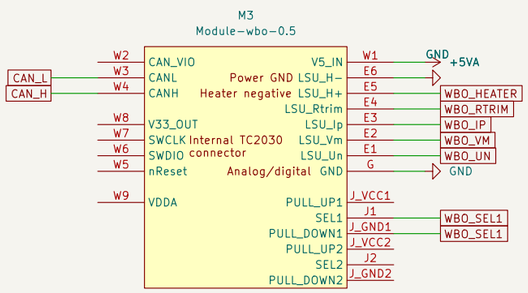
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File: INJ4.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 13/14

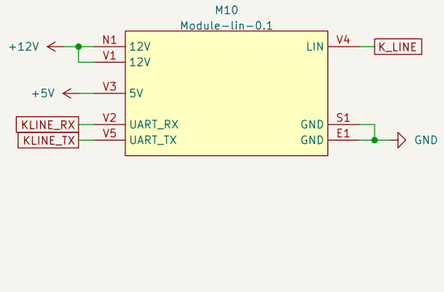


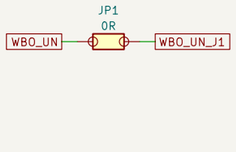
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File: INJ4.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 14/14

Communications.

M3: rusEFI wideband (WBO) for a Bosch LSU 4.9 front O2, on CAN to M1. It reaches the sensor through the stock harness (A33/A26/A25, heater A2+A3) via the 0R links, or through a pigtail on the pads (adapter table in README).
M10: LIN transceiver used as the K-line (SSM, D20). D29 reaches the K-line only through an open link.
CAN H/L on pads, with an optional 120R terminator for an external bus end.











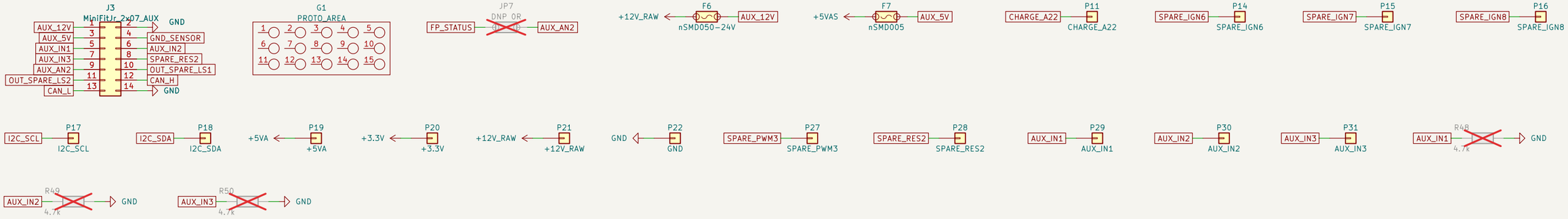
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File: Comms.kicad_sch

Title: Comms

Size: A3	Date: 2025-02-17	**Rev:**
KiCad E.D.A. 10.0.7		Id: 8/14

Aux connector and expansion.

J3: Molex Mini-Fit Jr 2x7 (pinout in documents/README.md): fused 12 V and 5 V, sensor ground, three digital inputs, two analog inputs, two low-side outputs, CAN.
Spare MCU pins (IGN6-8, I2C, OUT_PWM3, IN_RES2) and every rail are on pads next to the prototype area G1.



Sheet: /Aux and expansion/
File: Aux.kicad_sch

Title: Aux and expansion

Size: A3	Date: 2025-02-17	**Rev:**
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